Discovery of A Perceptual Distance Function for Measuring Image Similarity

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Abstract

For more than a decade, researchers have actively explored the area of image/video analysis and retrieval. Yet one fundamental problem remains largely unsolved: how to measure perceptual similarity between two objects. For this purpose, most researchers employ a Minkowski-type metric. Unfortunately, the Minkowski metric does not reliably find similarities in objects that are obviously alike. Through mining a large set of visual data, our team has discovered a perceptual distance function. We call the discovered function *dynamic partial distance function* (DPF). When we empirically compare DPF to Minkowski-type distance functions in image retrieval and in video shot-transition detection using our image features, DPF performs significantly better. The effectiveness of DPF can be explained by *similarity theories* in cognitive psychology.

Keywords: data mining, image retrieval, perceptual distance function, similarity search.

1 Introduction

Research in content-based image/video retrieval has steadily gained momentum in recent years as a result of the dramatic increase in the volume of digital images and videos. To achieve effective retrieval, an image/video system must be able to accurately characterize and quantify perceptual similarity. However, a fundamental challenge—how to measure perceptual similarity—remains largely unanswered. Various distance functions, such as the Minkowski metric [16], histogram Cosine distance [25], and fuzzy logic [9], have been used to measure similarity between feature vectors representing images (and hence video frames). Unfortunately, our experiments show that they frequently overlook obviously similar objects and hence are not adequate for measuring perceptual similarity.

Quantifying perceptual similarity is a difficult problem. Indeed, we may be decades away from fully understanding how human perception works. In our present project, we have mined

visual data extensively to discover a good perceptual distance function for measuring perceptual similarity. Our mining hypothesis is this: Suppose most of the similar images can be clustered in a feature space. We can then claim with high confidence that **1)** the feature space can adequately capture visual perception, and **2)** the distance function used for clustering images in that feature space can accurately model perceptual similarity.

We perform our mining operation in two stages. In the first stage, we isolate the distance function factor (we use the Euclidean distance) to find a reasonable feature set. In the second stage, we freeze the features to discover a perceptual distance function that can better cluster similar images in the feature space. In other words, our goal is to find a function that can keep similar images close together in the feature space, and at the same time, keep dissimilar images away. We call the discovered function *dynamic partial distance function* (DPF). When we empirically compare DPF to Minkowski-type distance functions in both image retrieval and video shot-transition detection, DPF performs markedly better.

Similarity is one of the central theoretical constructs in psychology [12, 23], probably related to human survival instincts. We believe that being able to quantify similarity accurately must also hold a central place in theories of information retrieval. Our excitement in discovering DPF does not arise merely from the practical effectiveness we found in two applications. More importantly, we find that DPF has roots in cognitive psychology. In Section 5, we will discuss the links between DPF and some *similarity theories* in cognitive psychology. These links not only help us understand why DPF works effectively, but they also provide clues for further improving its effectiveness.

In summary, the contributions of this paper are as follows:

1. We construct a mining dataset to find a feature set that can adequately represent images. In that feature space, we find distinct patterns of similar and dissimilar images, which lead to the discovery of DPF.

2. We derive DPF based on the observed patterns, and we provide methods for finding the optimal settings for the function's parameters.

3. Through empirical study, we demonstrate that DPF is very effective in finding images that have been transformed by rotation, scaling, downsampling, and cropping, as well as images that are perceptually similar to the query image.

4. Applying DPF to video shot-transition detection, we show that DPF is also more effective

than the Minkowski metric.

The rest of this paper is organized into four sections. Section 2 describes the construction of the mining dataset and the selection of a representative image feature set. In Section 3, we show how, using the mining results, we derive DPF and provide methods to find its optimal setting. The experimental results in Section 4 demonstrate that DPF significantly outperforms Minkowski-like metrics in finding similar images and in detecting video shot transitions using our image features. Finally, we present related work and explain the links between DPF and cognitive psychology in Section 5.

2 Mining Image Feature Set

This section depicts how the mining dataset was constructed in three steps: testbed setup (Section 2.1), feature extraction (Section 2.2), and feature selection (Section 2.3).

2.1 Image Testbed Setup

To ensure that sound inferences can be drawn from our mining results, we carefully construct the dataset. First, we prepare for a dataset that is comprehensive enough to cover a diversified set of images. To achieve this goal, we collect 60,000 JPEG images from Corel CDs and from the Internet. Second, we define "similarity" in a slightly restrictive way so that individuals' subjectivity can be excluded.[1] For each image in the 60,000-image set, we perform 24 transformations (described shortly), and hence form 60,000 similar-image sets. The total number of images in the testbed is 1.5 million.

The 24 image transformations we perform include the following:

1. Scaling.
   - Scale up then down. We scale each image up by 4 and 16 times, respectively, and then scale it back to the original size.
   - Scale down then up. We scale each image down by factors of 2, 4, and 8, respectively, then scale it back to the original size.

[1] We have considered adding images taken under different lighting conditions or with different camera parameters. We decided not to include them because they cannot be automatically generated from an image. Nevertheless, our experimental results (see Section 4.1) show that the perceptual distance function discovered during the mining process can be used effectively to find other perceptually similar images. In other words, our testbed consists of a good representation of similar images, and the mining results (i.e., training results) can be generalized to testing data consisting of perceptually similar images produced by other methods.

2. Downsampling. We downsample each image by seven different percentages: 10%, 20%, 30%, 40%, 50%, 70%, and 90%.

3. Cropping. We evenly remove the outer borders to reduce each image by 5%, 10%, 20%, 30%, 40%, 50%, 60%, and 70%, respectively, and then scale it back up to the original size.

4. Rotation. We rotate each image by 90, 180, and 270 degrees.

5. Format transformation. We obtain the GIF version of each JPEG image.

2.2 Image Features

To describe images, we must find a set of features that can represent those images adequately. Finding a universal representative feature set can be very challenging, since different imaging applications may require different feature sets. For instance, the feature set that is suitable for finding tumors may not be effective for finding landscape images, and vice versa. However, we believe that by carefully separating perception from intelligence (i.e., domain knowledge), we can identify meaningful perceptual features independent of their imaging applications.

Psychologists and physiologists divide the human visual system into two parts: the *perceiving part*, and the *inference part* [24]. The perceiving part receives photons, converts electrical signals into neuro-chemical signals, and delivers the signals to our brains. The inference part then analyzes the perceived data based on our knowledge and experience. A baby and an adult have equal capability for perceiving, but differing capability for understanding what is perceived. Among adults, specially trained ones can interpret an X-ray film, but the untrained cannot. In short, the perceiving part of our visual system is task-independent, so it can be characterized in a domain-independent manner.

We extract features such as color, shape, and texture from images. In the color channel, we characterize color in multiple resolutions. We first divide color into 12 color bins including 11 bins for culture colors and one bin for outliers [5]. At the coarsest resolution, we characterize color using a color mask of 12 bits. To record color information at finer resolutions, we record nine additional features for each color. These nine features are color histograms, color means (in H, S and V channels), color variances (in H, S and V channels), and two shape characteristics: elongation and spreadness. Color elongation characterizes the shape of a color, and spreadness characterizes how that color scatters within the image [8]. Table 1 summarizes color features in coarse, medium and fine resolutions.

Filter Name	Resolution	Representation
Masks	Coarse	Appearance of culture colors
Spread	Coarse	Spatial concentration of a color
Elongation	Coarse	Shape of a color
Histograms	Medium	Distribution of colors
Average	Medium	Similarity comparison within the same culture color
Variance	Fine	Similarity comparison within the same culture color

Table 1: Multi-resolution Color Features.

Texture is an important characteristic for image analysis. Studies [10, 11, 19, 21] have shown that characterizing texture features in terms of structuredness, orientation, and scale (coarseness) fits well with models of human perception. From the wide variety of texture analysis methods proposed in the past, we choose a discrete wavelet transformation (DWT) using quadrature mirror filters [19] because of its computational efficiency.

Each wavelet decomposition on a 2-D image yields four subimages: a $\frac{1}{2} \times \frac{1}{2}$ scaled-down image of the input image and its wavelets in three orientations: horizontal, vertical and diagonal. Decomposing the scaled-down image further, we obtain the tree-structured or wavelet packet decomposition. The wavelet image decomposition provides a representation that is easy to interpret. Every subimage contains information of a specific scale and orientation and also retains spatial information. We obtain nine texture combinations from subimages of three scales and three orientations. Since each subimage retains the spatial information of texture, we also compute elongation and spreadness for each texture channel. Figure 1 summarizes texture features.

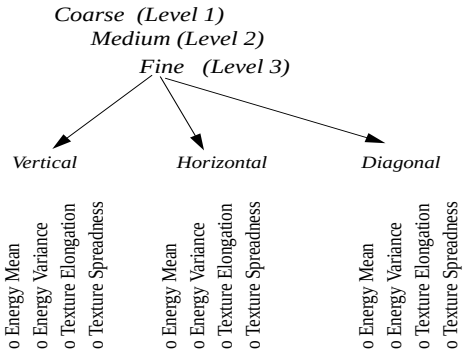


Figure 1: Multi-resolution Texture Features.

2.3 Feature Selection

Once the testbed is set up and relevant features extracted, we fix the distance function to examine various feature combinations. For the time being, we employ the Euclidean distance function to quantify the similarity between two feature vectors. We use the Euclidean function because it is commonly used, and it achieves acceptable results. (However, we will offer a replacement distance function for the Euclidean distance in Section 3.)

Using different feature combinations, we employ the Euclidean function to find the distance rankings of the 24 images that are similar to the original image (i.e., the query image). If a feature set can adequately capture the characteristics of images, the 24 similar images should be among those closest to the query image. (In an ideal case, the 24 similar images should be the 24 images closest to the query image.)

Our experiments reveal that when only individual features (e.g., color histograms, color elongation, and color spreadness) are employed, the distance function cannot easily capture the similar images even among the top-100 nearest neighbors. For a top-100 query, all individual features suffer from a dismal recall lower than 30%. When we combine all color features, the top-100 recall improves slightly, to 45%. When both color and texture features are used, the recall improves to 60%.

At this stage, we can go in either of two directions to improve recall. One, we can add more features, and two, we can replace the Euclidean distance function. We will consider adding additional features in our future work. In this paper, we focus on finding a perceptual distance function that improves upon the Euclidean Function.

3 Discovering the Dynamic Partial Distance Function

We first examine two most popular distance functions used for measuring image similarity: Minkowski function and weighted Minkowski function. Building upon those foundations, we explain the heuristics behind our new distance function—*Dynamic Partial Distance Function (DPF)*.

3.1 Minkowski Metric and Its Limitations

The Minkowski metric is widely used for measuring similarity between objects (e.g., images). Suppose two objects X and Y are represented by two p dimensional vectors $(x_1, x_2, \cdots, x_p)$ and

$(y_1, y_2, \cdots, y_p)$, respectively. The Minkowski metric $d(X, Y)$ is defined as

$$d(X, Y) = (\sum_{i=1}^{p} |x_i - y_i|^r)^{\frac{1}{r}}, \tag{1}$$

where r is the Minkowski factor for the norm. Particularly, when r is set as 2, it is the well known Euclidean distance; when r is 1, it is the Manhattan distance (or L_1 distance). An object located a smaller distance from a query object is deemed more similar to the query object. Measuring similarity by the Minkowski metric is based on one assumption: that similar objects should be similar to the query object in all dimensions.

A variant of the Minkowski function, the weighted Minkowski distance function, has also been applied to measure image similarity. The basic idea is to introduce weighting to identify important features. Assigning each feature a weighting coefficient w_i $(i = 1 \cdots p)$, the weighted Minkowski distance function is defined as:

$$d_w(X, Y) = (\sum_{i=1}^{p} w_i |x_i - y_i|^r)^{\frac{1}{r}}. \tag{2}$$

By applying a static weighting vector for measuring similarity, the weighted Minkowski distance function assumes that similar images resemble the query images in the same features. For example, when the function weights color features high and ignores texture features, this same weighting is applied to all pair-wise distance computation with the query image. We will show shortly that this *fixed* weighting method is restrictive in finding similar objects of different kinds.

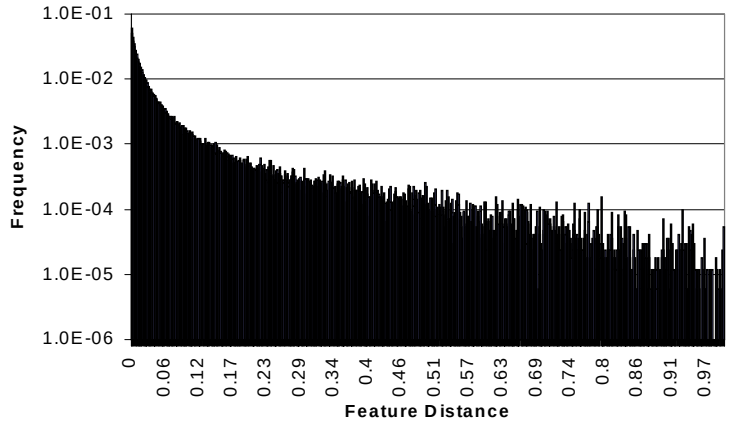
We questioned the above assumptions upon observing how similar objects are located in the feature space. For this purpose, we carried out extensive data mining work on a 1.5M-image dataset introduced in Section 2. To better discuss our findings, we introduce a term we have found useful in our data mining work. We define the *feature distance* on the ith feature as

$$\delta_i = |x_i - y_i|. \qquad (i = 1, \cdots, p)$$

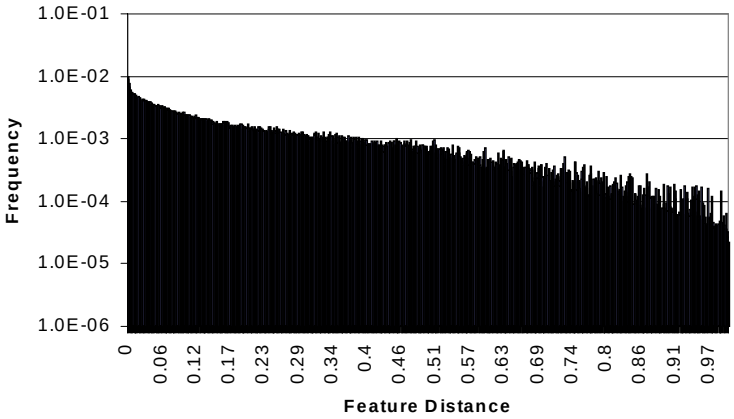
The expressions of Equation 1 and Equation 2 can be simplified into

$$d(X, Y) = (\sum_{i=1}^{p} {\delta_i}^r)^{\frac{1}{r}} \qquad and \qquad d_w(X, Y) = (\sum_{i=1}^{p} w_i {\delta_i}^r)^{\frac{1}{r}}.$$

In our mining work, we first tallied the feature distances between similar images (denoted as δ^+), and also those between dissimilar images (denoted as δ^-). Since we normalized feature



(a) Similar Images



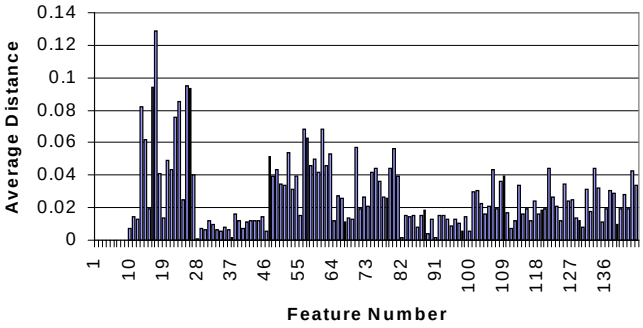
(b) Dissimilar Images

Figure 2: The Distributions of Feature Distances.

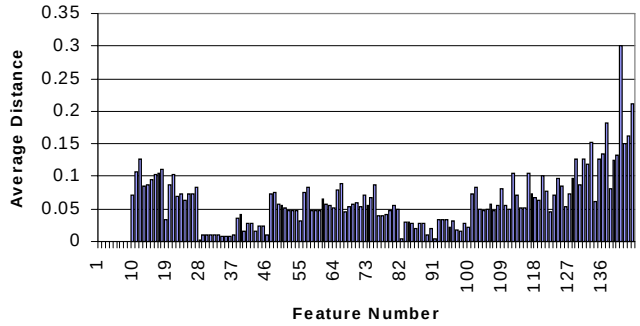
values to be between zero and one, the ranges of both δ^+ and δ^- are between zero and one. Figure 2 presents the distributions of δ^+ and δ^-. The x-axis shows the possible value of δ, from zero to one. The y-axis (in logarithmic scale) shows the percentage of the features at different δ values.

The figure shows that δ^+ and δ^- have different distribution patterns. The distribution of δ^+ is much skewed toward small values (Figure 2(a)), whereas the distribution of δ^- is more evenly distributed (Figure 2(b)). We can also see from Figure 2(a) that a moderate portion of δ^+ is in the high value range (≥ 0.5), which indicates that similar images may be quite dissimilar in some features. From this observation, we infer that the assumption of the Minkowski metric is inaccurate. Similar images are not necessarily similar in all features.

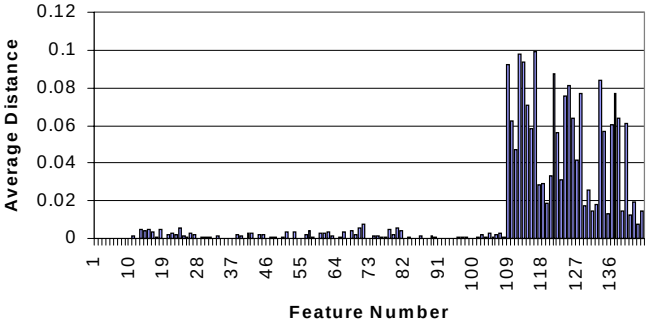
Furthermore, we examined whether similar images resemble the query images in the same



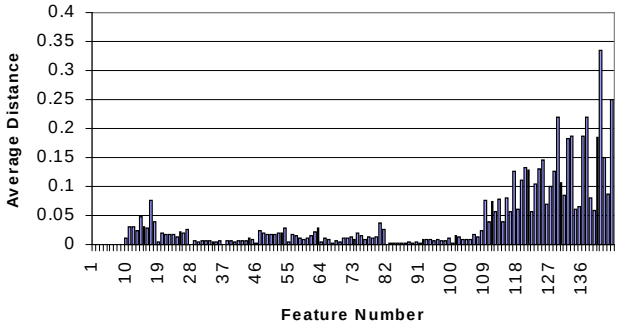
(a) GIF Images



(b) Cropped Images



(c) Rotated Images



(d) Scaled Images

Figure 3: The Average Feature Distances.

way. We tallied the feature distance (δ^+) of the 144 features for different kinds of image transformations. Figure 3 presents four representative transformations: GIF, cropped, rotated, and scaled. The x-axis of the figure depicts the feature numbers, from 1 to 144. The first 108 features are various color features, and the last 36 are texture features. The figure shows that various similar images can resemble the query images in very different ways. GIF images have larger δ^+ in color features (the first 108 features) than in texture features (the last 36 features). In contrast, cropped images have larger δ^+ in texture features than in color features. For rotated images, the δ^+ in colors comes close to zero, although its texture feature distance is much greater. A similar pattern appears in the scaled and the rotated images. However, the magnitude of the δ^+ of scaled images is very different from that of rotated images.

Our observations show that the assumptions made by the Minkowski and weighted Minkowski function are questionable.

1. Similar images do not resemble the query images in all features. Figure 2 shows that similar images different from a query image in many respects.

2. Images similar to the query images can be similar in differing features. Figure 3 shows that some images resemble the query image in texture, others in color.

The above observations not only refute the assumptions of Minkowski-type distance functions, but also provide hints as to how a good distance function would work. The first point is that a distance function does not need to consider all features equally, since similar images may match only some features of the query images. The second point is that a distance function should weight features dynamically, since various similar images may resemble the query image in differing ways. These points lead to the design of the *dynamic partial* distance function.

3.2 Dynamic Partial Distance Function

Based on the observations explained above, we designed a distance function to better represent the perceptual similarity. Let $\delta_i = |x_i - y_i|$, for $i = 1, \cdots, p$. We first define sets Δ_m as

$$\Delta_m = \{The \;\; smallest \;\; m \; \delta's \;\; of \;\; (\delta_1, ..., \delta_p)\}.$$

Then we define the *Dynamic Partial Distance Function* (DPF) as

$$d(m, r) = (\sum_{\delta_i \in \Delta_m} {\delta_i}^r)^{\frac{1}{r}}. \tag{3}$$

DPF has two adjustable parameters: m and r. Parameter m can range from 1 to p. When $m = p$, it degenerates to the Minkowski metric. When $m < p$, it counts only the smallest m feature distances between two objects, and the influence of the $(p - m)$ largest feature distances is eliminated. Note that DPF dynamically selects features to be considered for different pairs of objects. This is achieved by the introduction of Δ_m, which changes dynamically for different pairs of objects. In Section 4.1, we will show that if a proper value of m is chosen, it is possible to make similar images aggregate more compactly and locate closer to the query images, simultaneously keeping the dissimilar images away from the query images. In other words, similar and dissimilar images are better separated by DPF than by earlier methods.

The idea employed by DPF can also be generalized to improve the weighted Minkowski distance function. We modify the weighted Minkowski distance by defining the weighted DPF as

$$d_w(m, r) = (\sum_{\delta_i \in \Delta_m} w_i {\delta_i}^r)^{\frac{1}{r}}. \tag{4}$$

4 Empirical Study

We conducted an empirical study to examine the effectiveness of DPF. Our experiments consisted of three parts.

1. We compared DPF with the Euclidean distance function and L_1 distance function, the most widely used similarity functions in image retrieval. We also compared DPF with the histogram Cosine [2] distance function, which is also commonly used in information retrieval [20, 25]. (Section 4.1).

2. We tested whether DPF can be generalized to video shot-transition detection, the foundation of video analysis and retrieval applications (Section 4.2).

3. In addition to the unweighted versions, we also examined whether the weighted DPF is effective for enhancing the performance of the weighted Minkowski distance function

[2] The Cosine metric computes the direction difference between two feature vectors. Specifically, given two feature vectors $\mathbf{x}$ and $\mathbf{y}$, the Cosine metric is given as

$$D = 1 - \frac{\mathbf{x}^T \mathbf{y}}{|\mathbf{x}||\mathbf{y}|}$$

(Section 4.3).

4.1 Image Retrieval

Our empirical study of image retrieval consisted of two parts: training and testing. In the training part, we used the 1.5M-image dataset to predict the optimal m value for DPF. In the testing part, we set DPF with the optimal m value, and tested it on an independently constructed 50K-image dataset to examine its effectiveness.

4.1.1 Predicting m Through Training

The design goal of DPF is to better separate similar images from dissimilar ones. To meet this design goal, we must judiciously select parameter m. (We take the Euclidean distance function as the baseline, thus we set $r = 2$ for both DPF and the Minkowski distance function.)

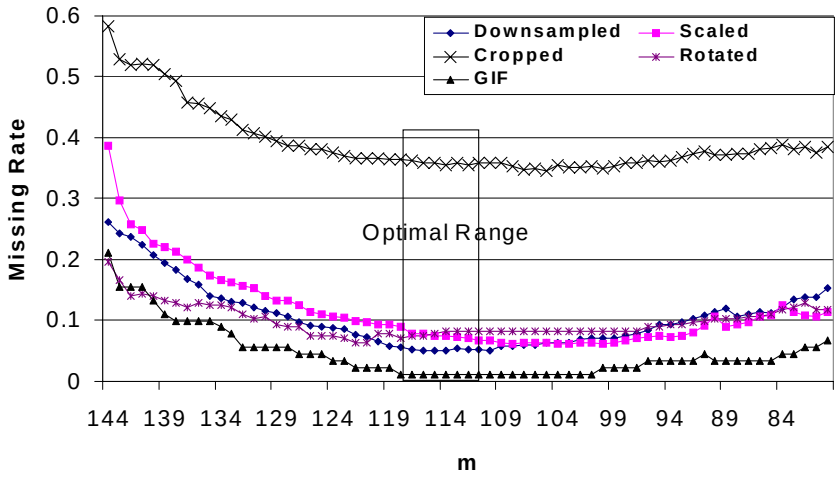
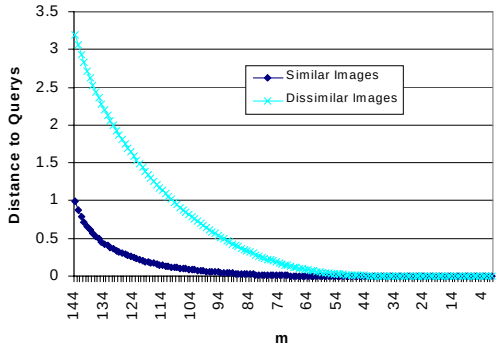


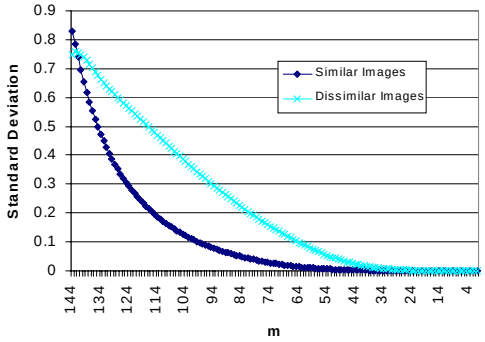
Figure 4: Training for the Optimal m Value.

To find the optimal m value, we used the $60,000$ original images to perform queries. we applied DPF of different m values to the 1.5M-image dataset. The 24 images with the shortest distance from each query image were retrieved. For each of the five similar-image categories (i.e., GIF, cropped, downsampled, rotated, or scaled), we observed how many of them failed to appear in the top-24 results. Figure 4 presents the average rate of missed images for each similar-image category. The figure shows that when m is reduced from 144 to between 110 and 118, the rates of missing are near their minimum for all five similar-image categories. (Note that when $m = 144$, DPF degenerates into the Euclidean function.) DPF outperforms the Euclidean

distance function by significant margins for all similar-image categories.



(a) Average of Distances

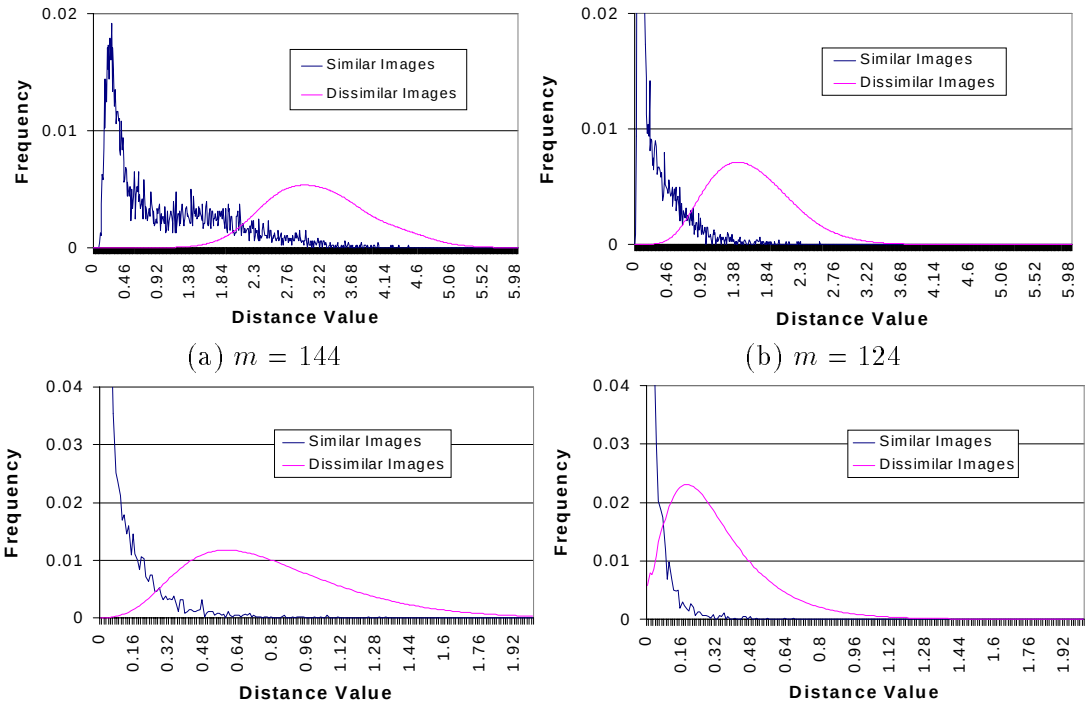


(b) Standard Deviation of Distances

Figure 5: The Effect of DPF.

To investigate why DPF works effectively when m is reduced, we tallied the distances from these $60,000$ queries to their similar images and their dissimilar images, respectively. We then computed the average and the standard deviation of these distances. We denote the average distance of the similar images to their queries as μ_d^+, of the dissimilar images as μ_d^-. We denote the standard deviation of the similar images' distances as σ_d^+, of the dissimilar images as σ_d^-.

Figure 5 depicts the effect of m (in the x-axis) on μ_d^+, μ_d^-, σ_d^+, and σ_d^-. Figure 5(a) shows that as m becomes smaller, both μ_d^+ and μ_d^- decrease. The average distance of similar images (μ_d^+), however, decreases at a faster pace than that of dissimilar images (μ_d^-). For instance, when we decrease m from 144 to 130, μ_d^+ decreases from 1.0 to about 0.3, a 70% decrease, whereas μ_d^- decreases from 3.2 to about 2.0, a 38% decrease. This gap indicates μ_d^+ is more sensitive to the m value than μ_d^-. Figure 5(b) shows that the standard deviations σ_d^+ and σ_d^- observe the same trend as the average distances do. When m decreases, similar images become more compact in the feature space at a faster pace than dissimilar images do.



(a) $m = 144$ (b) $m = 124$

(c) $m = 104$ (d) $m = 84$

Figure 6: Distance Distributions vs. m.

To provide more detailed information, Figure 6 depicts the distance distributions at four different m values. Figure 6(a) shows that when $m = 144$, a significant overlap occurs between the distance distributions of similar and dissimilar images to the query images. (When $m = 144$, DPF degenerates to the Euclidean function.) In other words, many similar images and dissimilar images may reside about the same distance from their query image, which causes degraded search performance. When we decrease m to 124, Figure 6(b) shows that both distributions shift toward the left. The distribution of similar images becomes more compact, and this leads to a better separation from dissimilar images. Further decreasing the m value moves both distributions leftward (as shown in Figures 6(c) and 6(d)). When little room is left for the distance distribution of similar images to move leftward, the overlap can eventually increase. Our observations from these figures confirm that we need to find the optimal m value to achieve best separation for similar and dissimilar images.

4.1.2 Testing DPF

We tested our distance functions on a dataset that was independently constructed from the 1.5M-image dataset used for conducting mining and parameter training.

The test dataset consisted of 50K randomly collected World Wide Web images. Among these images we identified 100 images as query images. For each query image, we generated 24 similar images using the transformation methods described in Section 2. We also visually identified 5 similar images for each query image. (See Figure 7 for examples of visually-identified similar images).



Figure 7: A Query Image (The Upper Left One) and Five Visually Identified Similar Images.

We conducted 100 queries using the 100 query images. For each query, we recorded the distance ranks of its similar images. For DPF, we fixed m value as 114 based on the training results in Section 4.1.1. Figure 8 depicts the experimental results. The precision-recall curves in the figure show that the search performance of DPF is significantly better than the other traditional distance functions. For instance, to achieve a recall of 80%, the retrieval precision of DPF is 84%, whereas the precision of the L_1 distance, the Euclidean distance, and the histogram Cosine distance is 70%, 50%, and 25%, respectively.

We were particularly interested in the retrieval performance of the visually identified similar images, which were not included into the training-image dataset. Figure 9 compares the retrieval performance of DPF and traditional distances for the visually identified similar images. The precision-recall curves indicate that, even though the visually identified similar images were not included in the training-image dataset, DPF could still find them effectively in the testing phase. This indicates that the trained DPF parameters can be generalized to find similar images produced by methods other than those for producing the training dataset.

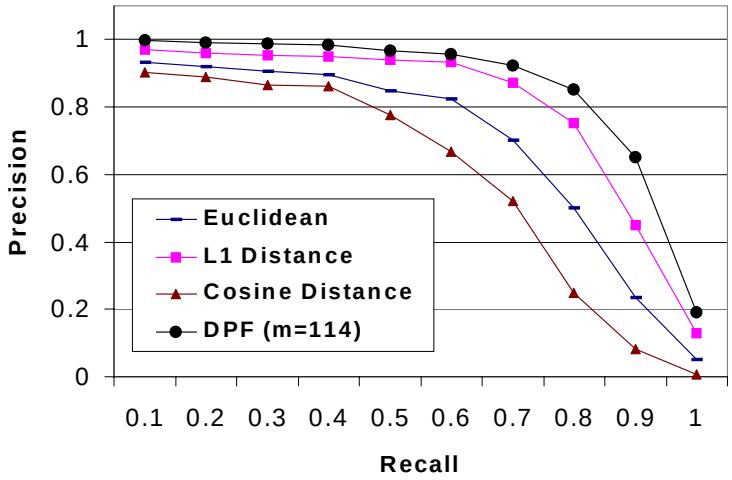


Figure 8: Precision/Recall for Similar Images.

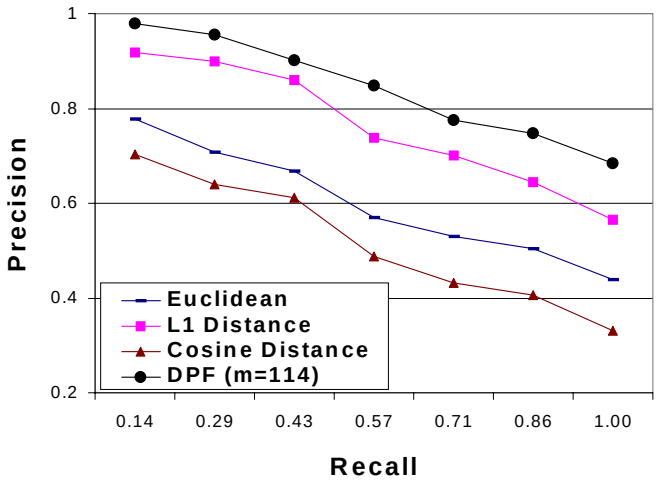


Figure 9: Precision/Recall for Visually Identified Similar Images.

4.2 Video Shot-Transition Detection

To further examine the generality of the DPF, we experimented DPF in another application—video shot-transition detection.

Our video dataset consisted of 150 video clips which contained thousands of shots. The videos covered following subjects:

- Cartoon: 30 clips, each clip lasting for 50 seconds (from commercial CDs).
- Comedy: 50 clips, each lasting for up to 30 seconds [1].

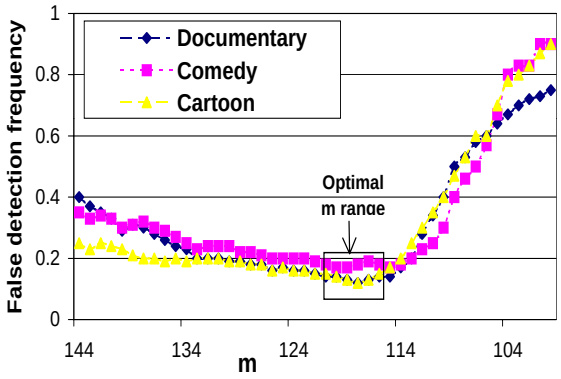


Figure 10: Optimal m.

• Documentary: 70 clips, each lasting for two to five minutes [2].

For characterizing a frame, we extracted the same set of 144 features for each frame, since these features can represent images to a reasonable extent. Our experiments had two goals. The first was to find the optimal parameter m settings for DPF (Section 4.2.1). The second was to compare the shot detection accuracy between employing DPF and employing the Minkowski metric as the inter-frame distance function (Section 4.2.2).

4.2.1 Parameter m

We fixed $r = 2$ in our empirical study. Then we took a machine learning approach to train the value of m. We sampled 40% of the video clips as the training data to discover a good m. We then used the remaining 60% of video clips as testing data to examine the effectiveness of the learned m.

In the training phase, we labeled the accurate positions of shot boundaries. We then experimented with different values of m on three video datasets (cartoon, comedy, and documentary). Figure 10 shows that for all three video types, the false detection rates are reduced to a minimum as m is reduced from 144 to between 115 and 120. (Recall that when $m = 144$, DPF degenerates into the Minkowski distance function.) It is evident that the Minkowski distance function is not the best choice for our purpose.

4.2.2 DPF vs. Minkowski

We next compared two inter-frame distance functions, DPF and Euclidean, on the testing data. For DPF, we set $m = 117$ based on the training results in Section 4.2.1. Table 2 shows that

DPF improves the detection accuracy over the Euclidean distance function on both precision and recall for all video categories. The average improvement as shown in Figure 11 is about 7% in both recall and precision. In other words, for every 100 shot transitions to be detected, DPF makes 7 fewer detection errors, a marked improvement.

Distance Functions	*Video Type*	Comedy	*Cartoon*	*Documentary*
	# of Shot Boundaries	425	167	793
Euclidean	*# of false*	93	39	192
	# of miss	97	37	183
	Precision	78.1%	76.6%	75.8%
	Recall	77.2%	77.8%	76.9%
DPF	*# of false*	61	26	140
	# of miss	67	25	129
	Precision	85.6%	84.4%	82.3%
	Recall	84.2%	85.0%	83.7%

Table 2: Precision and Recall.

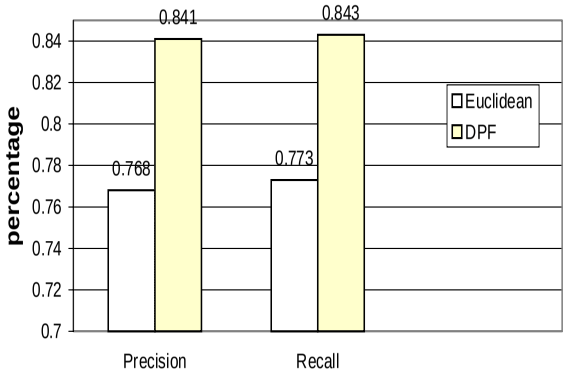
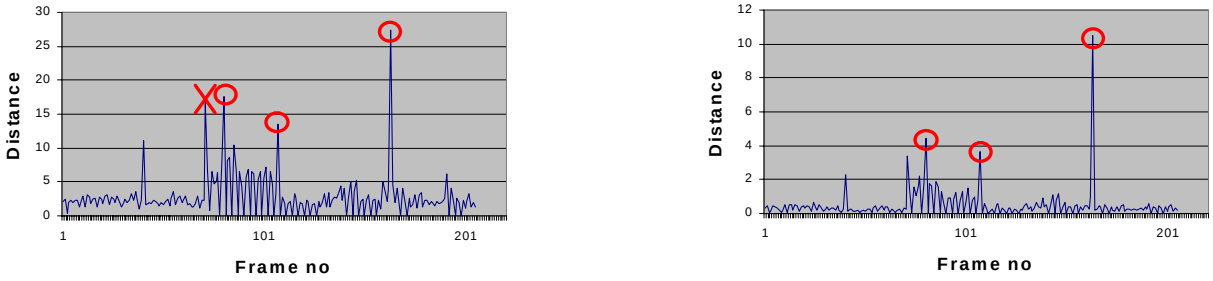


Figure 11: Overall Precision and Recall Comparison.

Figure 12 illustrates why DPF can better detect shot boundaries than Euclidean distance, from the signal/noise ratio perspective. The x-axis of the figure depicts the frame number; the y-axis depicts the inter-frame distance between the ith and the $(i+1)^{th}$ frames. We mark each real shot boundary with a circle and a false detection with a cross. Figure 12(a) shows that the Euclidean distance function identified four shot boundaries, in which the left-most one was a false positive. Figure 12(b) shows that DPF separates the distances between shot boundaries and non-boundaries better, and hence eliminates the one mis-detection. DPF improves the signal/noise ratio, and therefore, it is more effective in detecting shot transitions.



(a) Euclidean (b) DPF

Figure 12: Euclidean vs. DPF.

4.3 Weighted DPF vs. Weighted Euclidean

We were also interested in applying weighted DPF to improve the weighted Minkowski distance function, which has been used extensively to personalize similarity measures. For weighted Minkowski distance, a weighting vector is learned for each query. Usually, the weight of a feature is set as the inverse of the variance of its values among similar images. Here, we allowed the weighted Euclidean distance function to work under the ideal condition—that is, it knows all similar images a priori and can compute the ideal weighting vector for each query. Figure 13 shows that the weighted Euclidean function outperforms its unweighted counterpart. This result confirms that the weighted version [15, 14] is indeed a better choice than the unweighted version (provided that the appropriate weighting can be learned). However, there is still much room for improvement. When we applied weighted DPF using the same weighting vector, its retrieval performance was better than that of the weighted Euclidean distance function. For instance, at 80% recall rate, the retrieval precision of the weighted Euclidean distance is about 68%, whereas the weighted DPF could achieve a precision of above 85%. Again, our empirical study shows that the generalized form of DPF, weighted DPF, can be used to markedly enhance the weighted Minkowski distance for measuring image similarity.

4.4 Observations

We summarize the results of our experiments as follows:

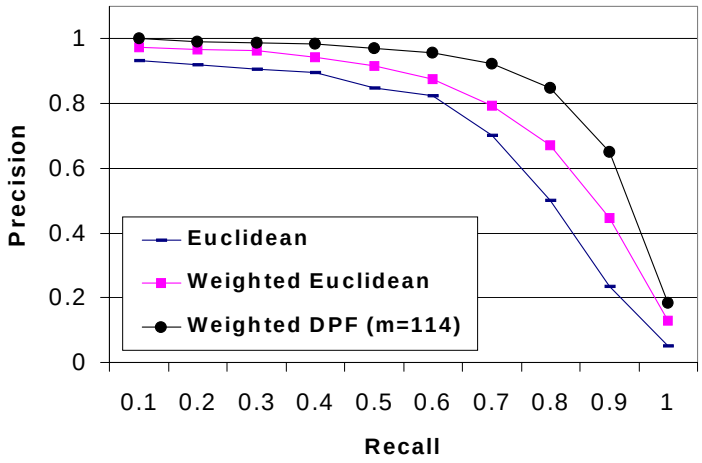


Figure 13: Comparision of Weighted Functions.

1. DPF is more effective than some most representative distance functions used in the CBIR community (e.g., Minkowski-like and histogram Cosine distance functions) for measuring image similarity and for detecting shot transitions.

2. The weighted version of DPF outperforms the weighted version of the Euclidean distance function.

3. We believe that DPF can be generalized to find similar images of some other ways, and that DPF can be effective when a different set of low-level features are employed. Our belief is partially supported by our empirical results, and partially justified by similar theories in cognitive science, which we discuss next.

5 Related Work and Closing Remarks

Similarity is one of the most central theoretical constructs in psychology [12, 23]. Its also plays a central role in information categorization and retrieval. Here we summarize related work in similarity distance functions. Using our experimental results, together with theories and examples in cognitive psychology, we explain why DPF works effectively as we discuss the progress of the following three similarity paradigms in cognitive psychology.

1. *Similarity is a measure of all respects.* As we discussed in Section 3, a Minkowski-like metric accounts for all respects (i.e., all features) when it is employed to measure similarity between two objects. Our mining result shown in Figure 2 is just one of a large number of

counter-examples demonstrating that the assumption of the Minkowski-like metric is questionable. The psychology studies of [12, 23] present examples showing that the Minkowski model appears to violate human similarity judgements.

2. *Similarity is a measure of a fixed set of respects.* Substantial work on similarity has been carried out by cognitive psychologists. The most influential work is perhaps that of Tversky [23], who suggests that similarity is determined by matching features of compared objects, and integrating these features by the formula

$$S(A, B) = \theta f(A \cap B) - \alpha f(A - B) - \beta f(B - A). \tag{5}$$

The similarity of A to B, $S(A, B)$, is expressed as a linear combination of the common and distinct features. The term $(A \cap B)$ represents the common features of A and B. $(A - B)$ represents the features that A has but B does not; $(B - A)$ represents the features that B has but A does not. The terms θ, α, and β refect the weights given to the common and distinctive components, and function f is often assumed to be additive [12].

The weighted Minkowski function [17] and the quadratic-form distances [3, 6] are the two representative distance functions that match the spirit of Equation 5. The weights of the distance functions can be learned via techniques such as relevance feedback [15, 17], principal component analysis, and discriminative analysis [26]. Given some similar and some dissimilar objects, the weights can be adjusted so that similar objects can be better distinguished from other objects.

However, the assumption made by these distance functions, that all similar objects are similar in the same respects [26], is questionable. As we have shown in Figure 3, GIF, cropped, rotated, and scaled images are all similar to the original images, but in differing features. For this reason, DPF allows objects to be matched to the query in different respects dynamically.

3. *Similarity is a process that provides respects for measuring similarity.* Suppose we are asked to name two places that are similar to England. Among several possibilities, Scotland and New England could be two reasonable answers. However, the respects England is similar to Scotland differ from those in which England is similar to New England. If we use the shared attributes of England and Scotland to compare England and New England, the latter pair might not be similar, and vice versa. This example echos the result presented in Figure 3—that is, objects can be similar to the query object in different respects. A

distance function using a fixed set of respects cannot capture objects that are similar in different sets of respects.

Murphy and Medin [13] provide early insights into how similarity works in human perception: "The explanatory work is on the level of determining which attributes will be selected, with similarity being at least as much a consequence as a cause of a concept coherence." Goldstone [4] explains that similarity is the process that determines the respects for measuring similarity. In other words, a distance function for measuring a pair of objects is formulated only after the objects are compared, not before the comparison is made. The respects for the comparison are activated in this formulation process. The activated respects are more likely to be those that can support coherence between the compared objects.

With those paradigms in mind, let us re-examine how DPF works. DPF activates different features for different object pairs. The activated features are those with minimum differences– those which provide coherence between the objects. If coherence can be maintained (because sufficient a number of features are similar), then the objects paired are perceived as similar. Cognitive psychology seems able to explain much of the effectiveness of DPF.

We plan to extend our study in at least two directions. First, the activation of respects is believed to be context-sensitive [7, 18, 22]. Also, certain respects may be more salient than others, and hence additional weighting factors should be considered. Second, since DPF is a non-metric function, non-metric space clustering (for information categorization) and indexing (for information organization and retrieval) need to be investigated.

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